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ACME: A Body-Worn Multimodal Sensor Suite Enabling Autonomous Documentation of Prehospital Care

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Abstract (2190/2200 Characters)

Introduction

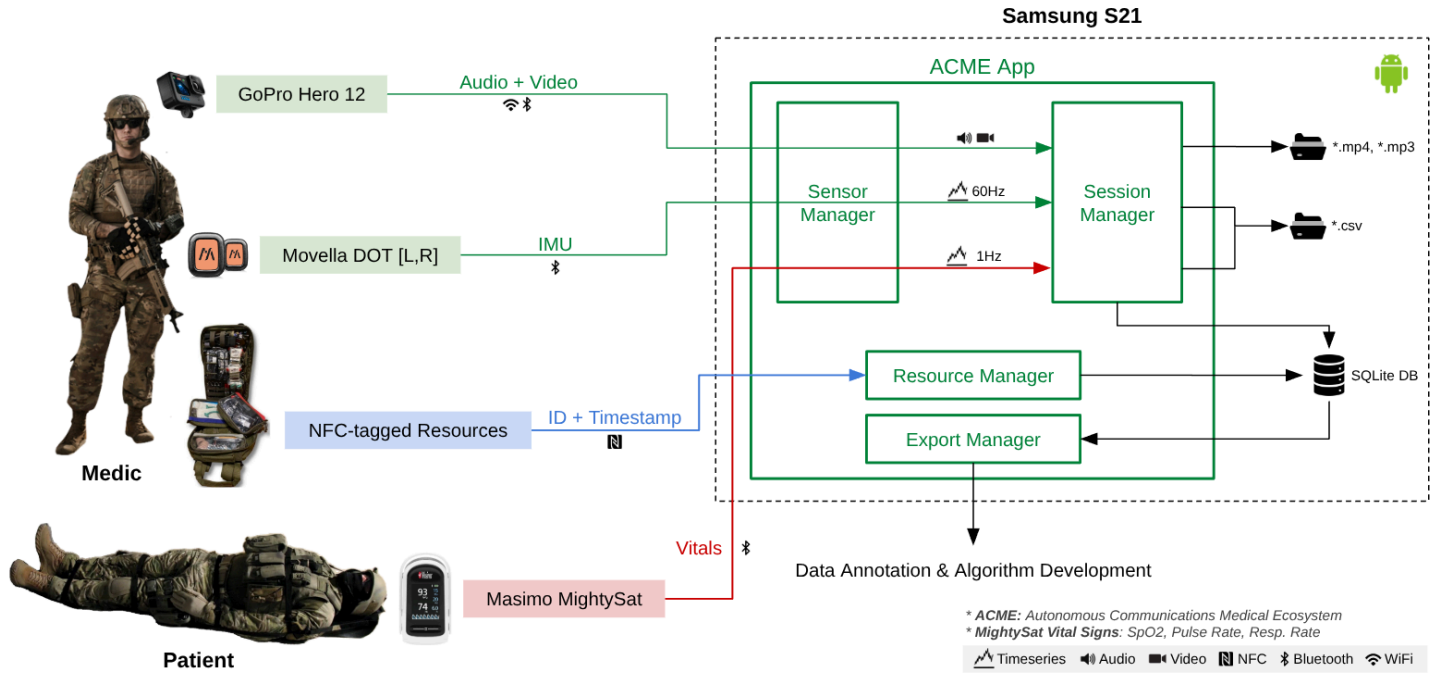
Manual documentation during time-critical prehospital care diverts attention from patient management and yields incomplete care records. Autonomous documentation requires a passive, multimodal data-capture system that records medic actions, patient physiology, and resource use without disrupting workflows. No integrated, field-ready platform currently exists.

Objectives

We aimed to build a body-worn sensor suite that 1) passively acquires time-synchronized video, audio, and motion data from the medic and patient vital signs; 2) minimizes setup burden via near-field communication (NFC) pairing; and 3) exports datasets for downstream task-identification and documentation algorithms.

Methods

We developed the Autonomous Communications Medical Ecosystem (ACME) as a native Android application running on a Samsung Galaxy S21. A GoPro Hero 12 provided 720p30 video/audio; bilateral Movella DOT inertial measurement units streamed acceleration/gyroscope data at 60 Hz; and a Masimo MightySat pulse-oximeter delivered SpO_2, pulse, and respiration rate at 1 Hz. Medical supplies were tagged with NFC stickers; tapping a tag against the phone logged item identity and timestamp. Sensors were paired via NFC tap-to-pair and controlled via the app. All streams were synchronized by timestamping to the Unix epoch and stored locally as CSV (IMU, vitals) and MP4/MP3 (video/audio).



ACME Architecture Diagram

Results

Continuous data collection from all sensors was sustained for up to 2.5 hours, limited by GoPro battery capacity. All data streams remained synchronized with the application clock without manual intervention. External evaluation by the U.S. Army's Telemedicine and Advanced Technology Research Center has deployed the platform in over 100 high-fidelity prehospital simulations to generate datasets for algorithm development; formal performance metrics from this evaluation are pending.

Conclusions

ACME demonstrates engineering feasibility for passive multimodal data capture in prehospital settings and provides a foundation for developing autonomous documentation pipelines. Next steps include real-time integration with task-identification algorithms and further testing in real-world scenarios.